

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



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Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Midterm

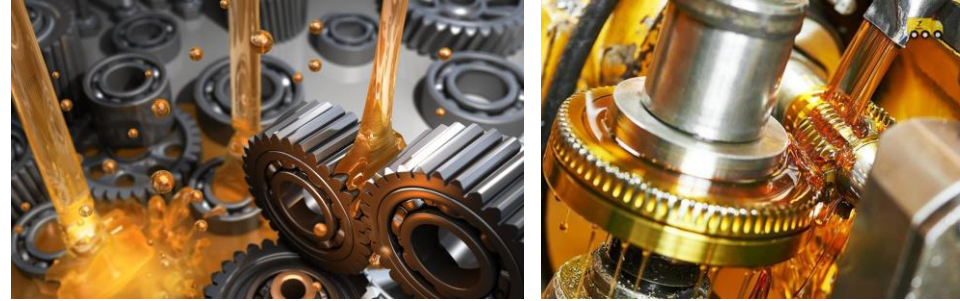
Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

What is Lubricating Oil?

Lubricating Oil (=Lubricant, Lub. Oil, L.O.)

A fluid that forms a thin film between metal surfaces to reduce friction and wear, remove heat, and help machines operate smoothly.



1. Lubrication

Forms an oil film between metal surfaces to prevent direct metal-to-metal contact.

(Reduces friction → prevents wear → extends machine life.)

2. Cooling

Absorbs frictional and combustion heat generated from engines and bearings and transfers it to other areas. (Prevents overheating and thermal deformation.)

What is Lubricating Oil?

3. Cleaning

Impurities such as wear particles, carbon, and sludge adhere to the oil film, circulate, and are removed through a filter.

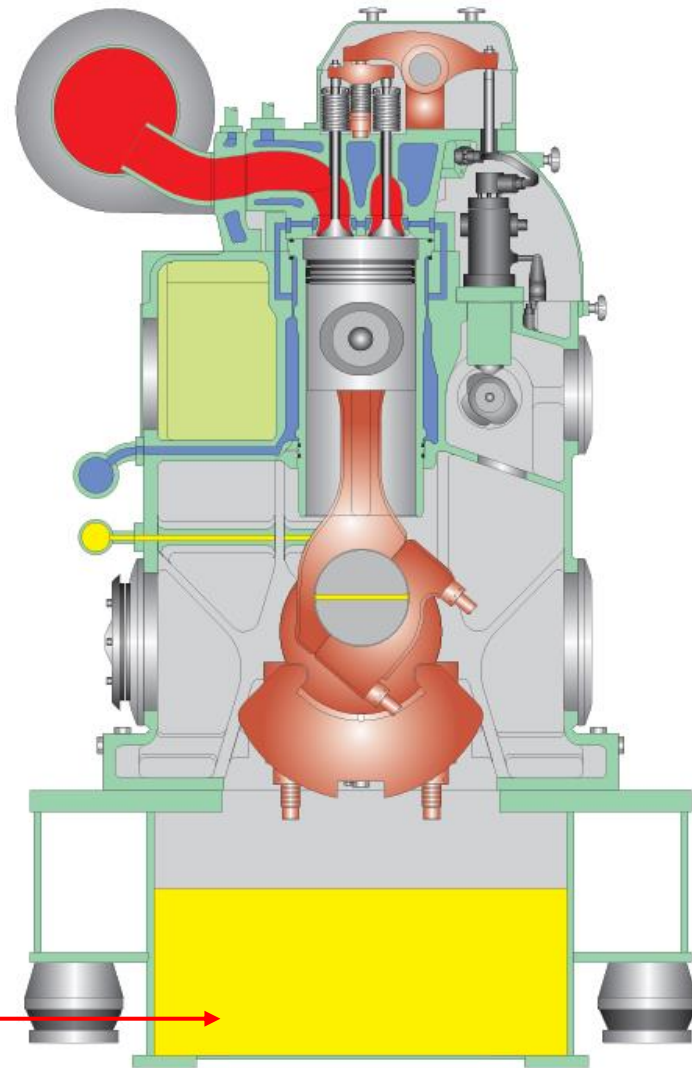
4. Sealing

Seals the gap between the piston ring and cylinder liner to prevent combustion gas leakage. Contributes to maintaining compression ratio and improving power output.

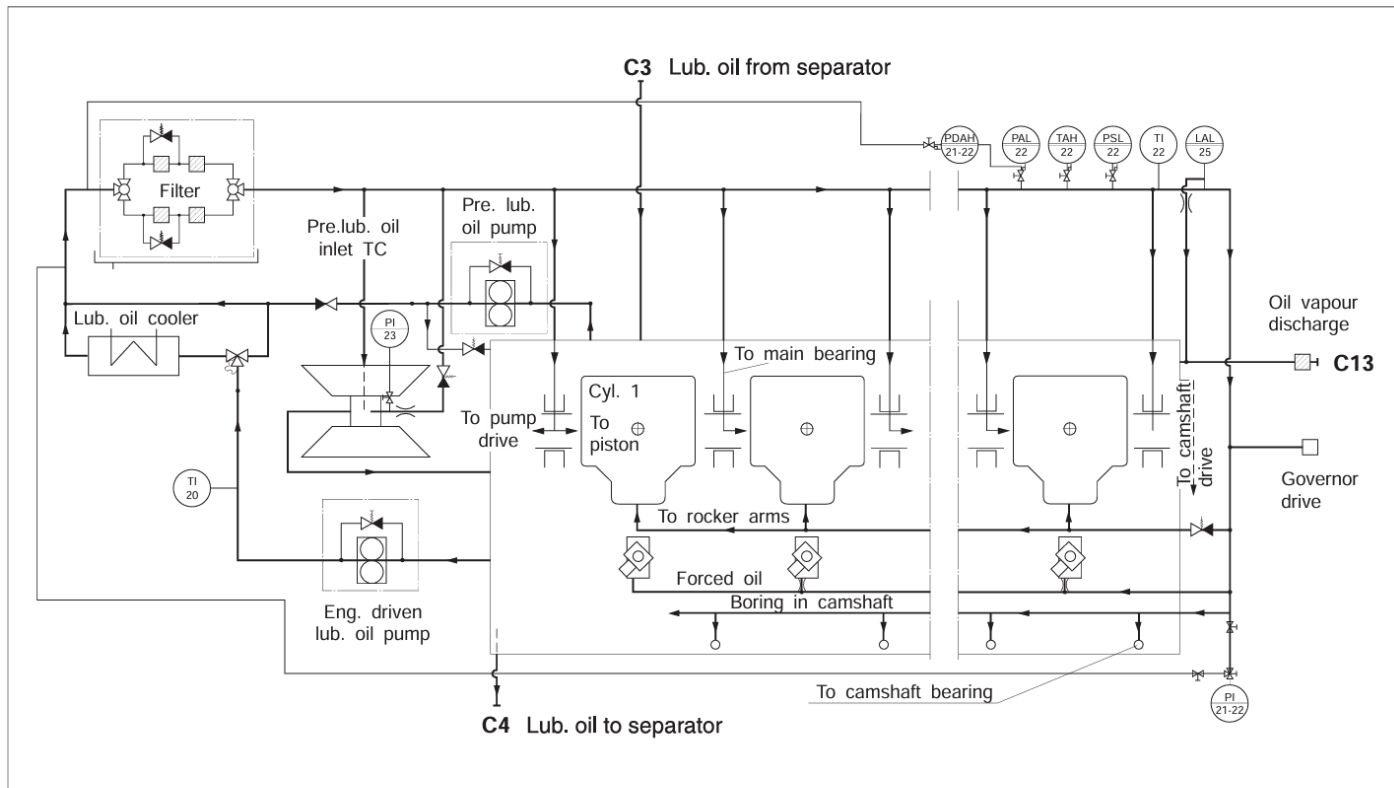
5. Corrosion & Rust Prevention

Forms a film on metal surfaces to protect them from oxygen, moisture, and combustion byproducts. Protection from acidic byproducts is particularly important in marine engines.

L.O in L.O sump tank



Internal Lubricating Oil System



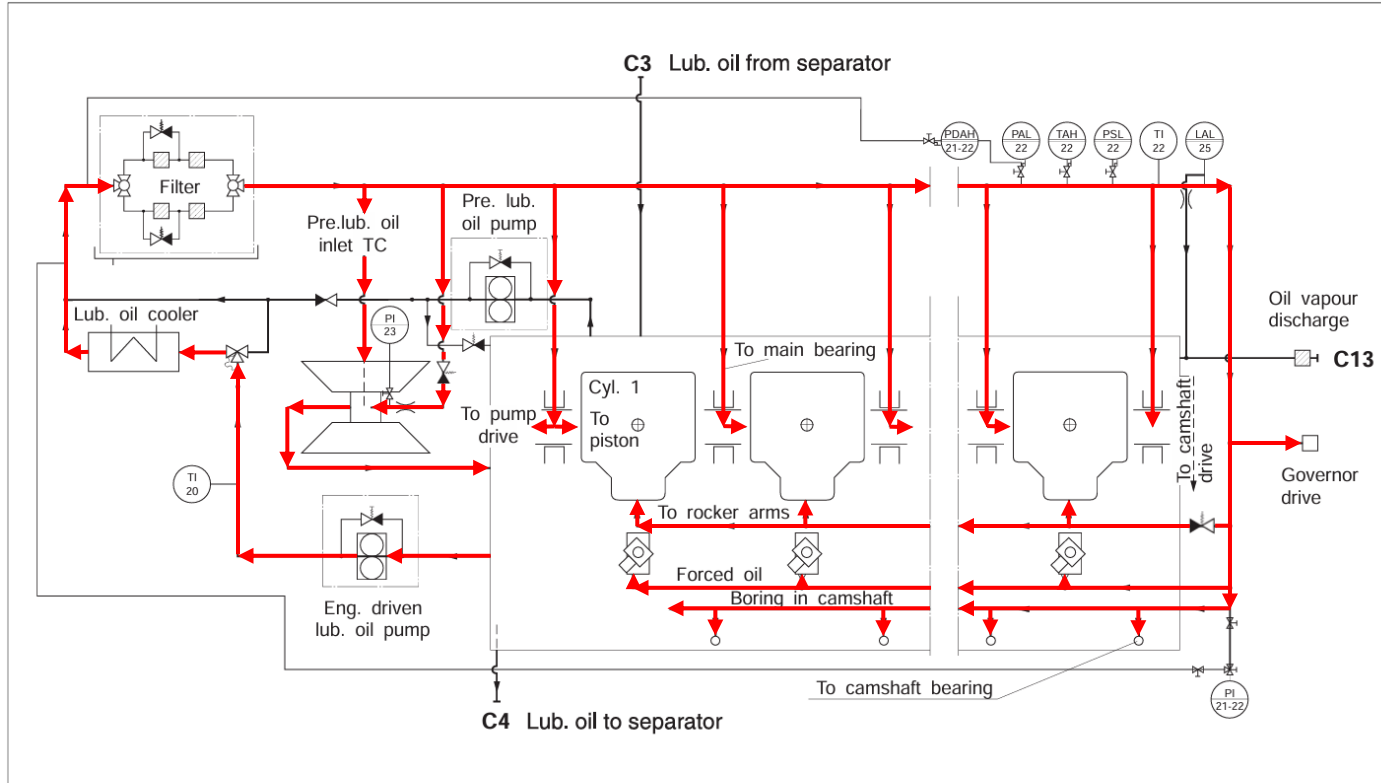
The lubricating oil system is based on wet sump lubrication.

All moving parts of the engine are lubricated with oil circulating under pressure in a closed built-on system.

The lubricating oil is furthermore used for the purpose of cooling the pistons.

Fig. 1. Diagram for Internal Lubricating Oil System

Internal Lubricating Oil System

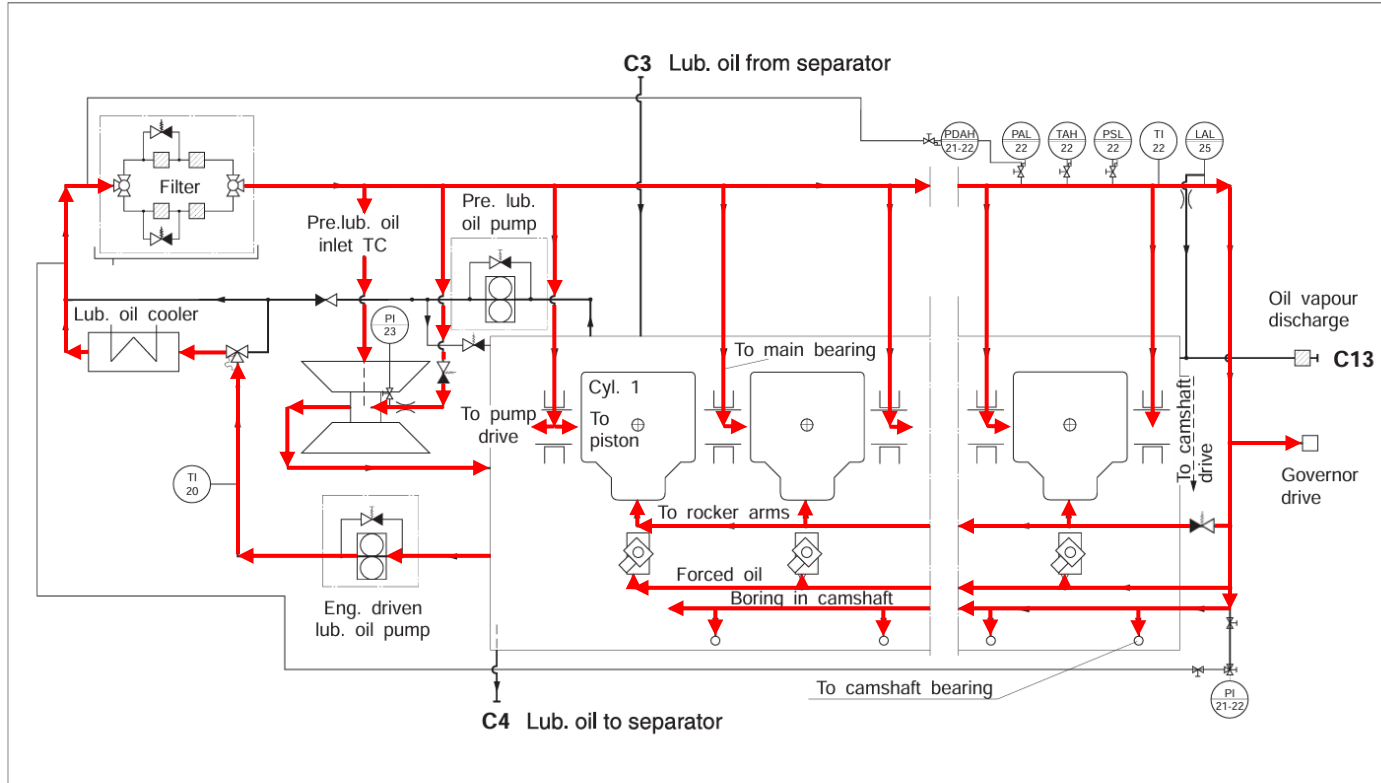


The lubricating oil pump draws oil from the oil sump and presses the oil through the cooler and filter to the main lubricating oil pipe, from where the oil is distributed to the individual lubricating points.

From the lubricating points the oil returns by gravity to the oil sump.

Fig. 1. Diagram for Internal Lubricating Oil System

Internal Lubricating Oil System



The main groups of components to be lubricated are:

- 1 - Turbocharger
- 2 - Main bearings, big-end bearings
- 3 - Camshaft drive
- 4 - Governor drive
- 5 - Rocker arms
- 6 - Camshaft

Fig. 1. Diagram for Internal Lubricating Oil System

Internal Lubricating Oil System

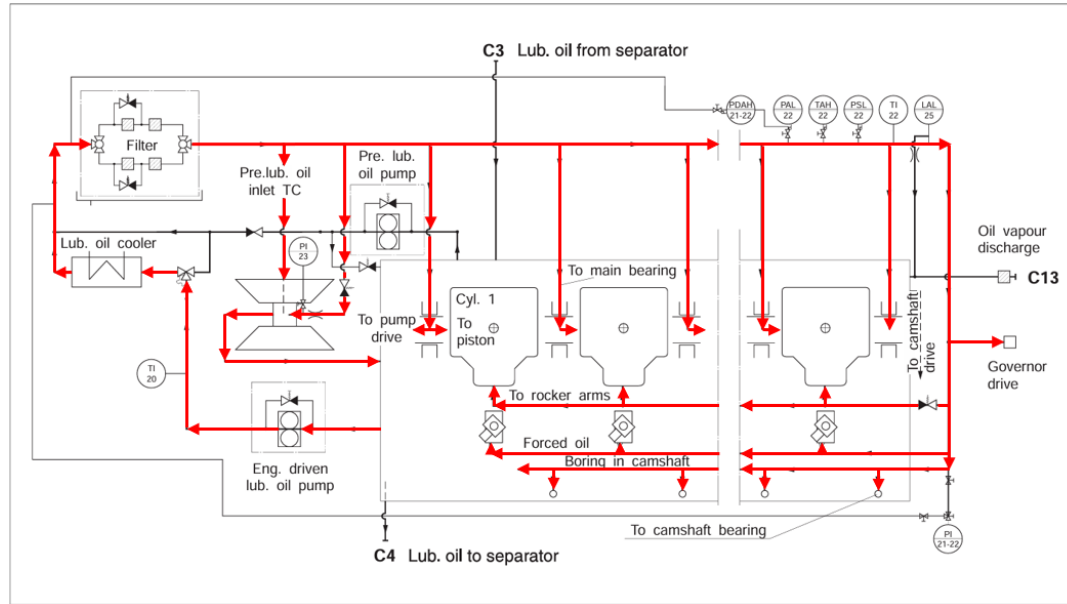


Fig. 1. Diagram for Internal Lubricating Oil System

For priming and during operation, the turbocharger is connected to the lub. oil circuit of the engine, the oil serves for bearing lubrication.

The inlet line to the turbocharger is equipped with a fixed throttle in order to adjust the oil flow and a non return valve to prevent draining during stand-still.

The non-return valve has back-pressure function requiring a pressure slightly above the priming pressure to open in normal flow direction. In this way overflowing of the turbocharger is prevented during stand-still periods, where the prelubricating pump is running.

Internal Lubricating Oil System

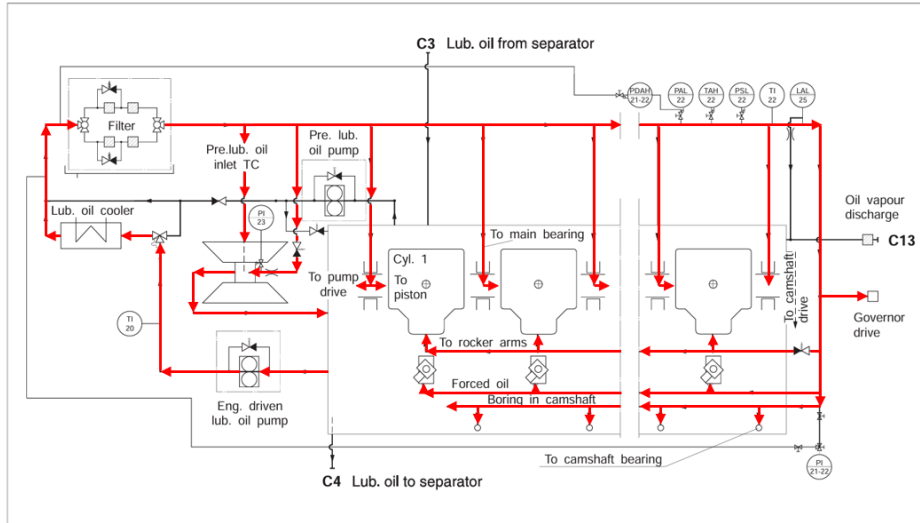
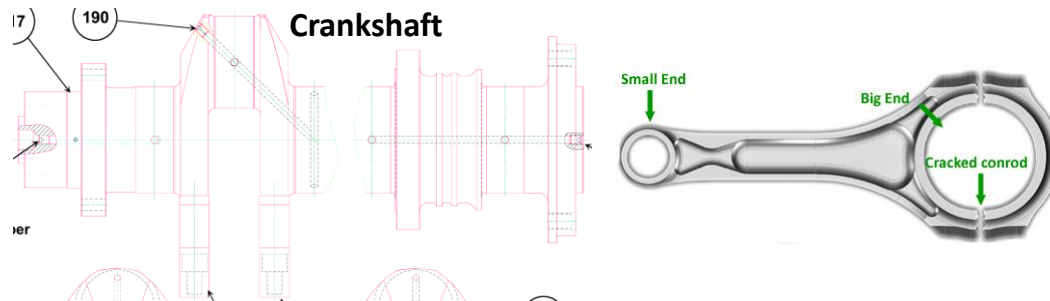


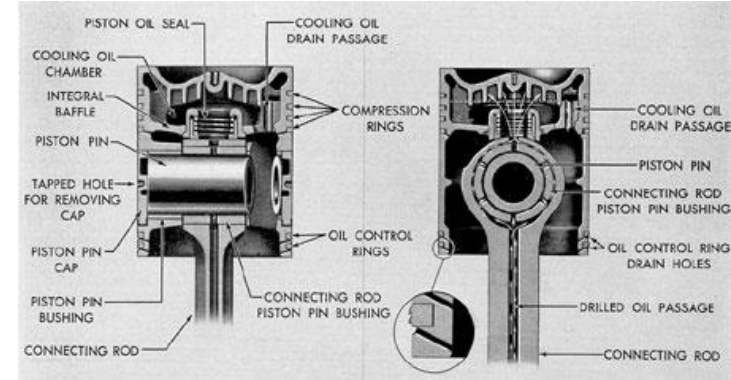
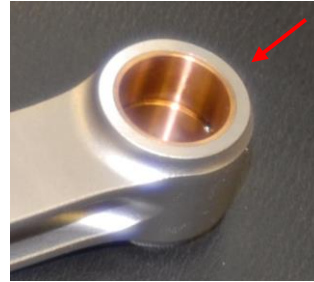
Fig. 1. Diagram for Internal Lubricating Oil System



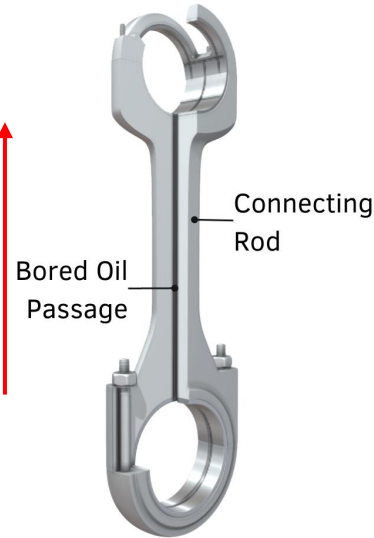
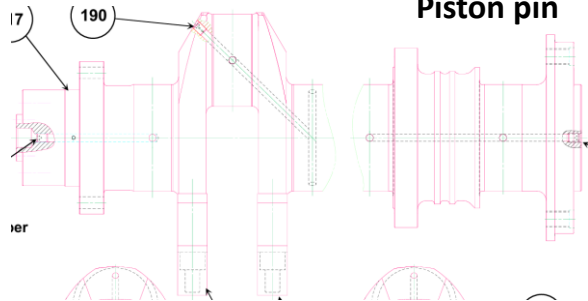
Lubricating oil for the main bearings is supplied through holes drilled in the engine frame. From the main bearings it passes through bores in the crankshaft to the connecting rod big-end bearings.

The connecting rods have bored channels for supply of oil from the big-end bearings to the small-end bearings, which has an inner circumferential groove, and a pocket for distribution of oil in the bush itself and for supply of oil to the pin bosses and the piston cooling through holes and channels in the piston pin.

Internal Lubricating Oil System



Piston pin



The connecting rods have bored channels for supply of oil from the big-end bearings to the small-end bearings, which has an inner circumferential groove, and a pocket for distribution of oil in the bush itself and for supply of oil to the pin bosses and the piston cooling through holes and channels in the piston pin.

Internal Lubricating Oil System

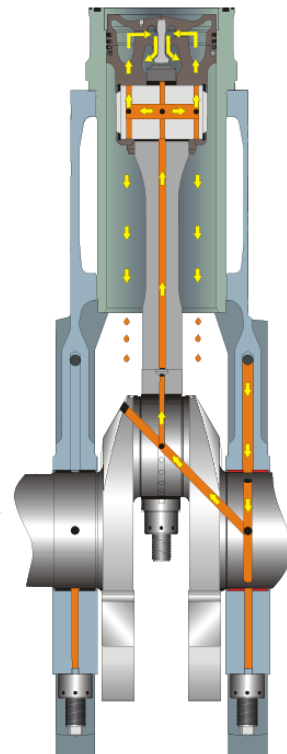
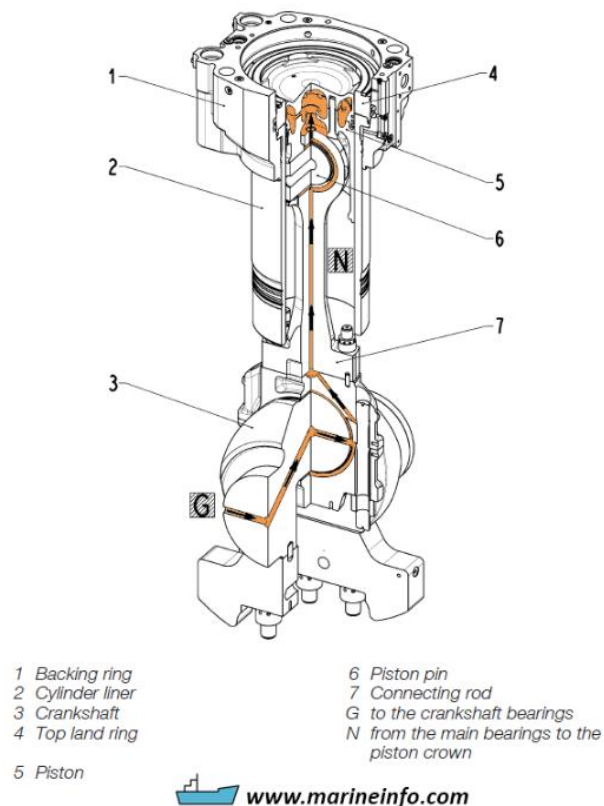


Fig.1 The lubricating oil flow in connecting rod and piston (커넥팅 로드 및 피스톤내의 윤활유 흐름)

Internal Lubricating Oil System

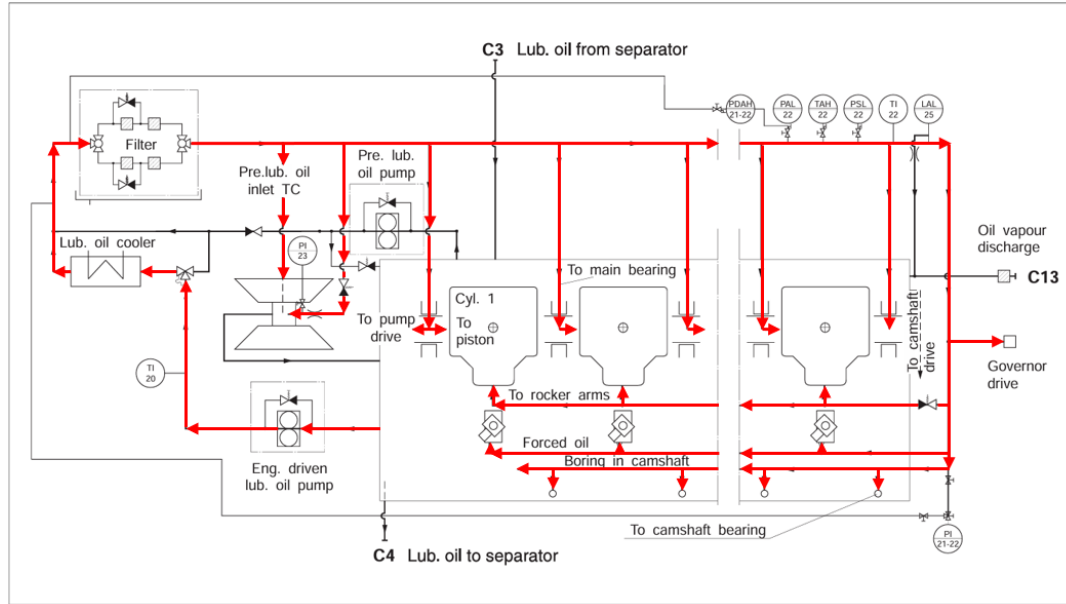
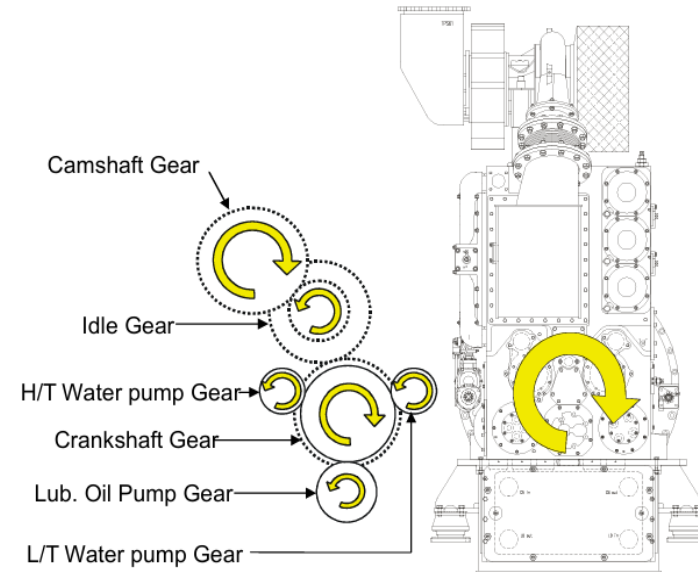


Fig. 1. Diagram for Internal Lubricating Oil System

From the front main bearings, channels are bored in the crankshaft for lubricating of the pump drive.



Internal Lubricating Oil System

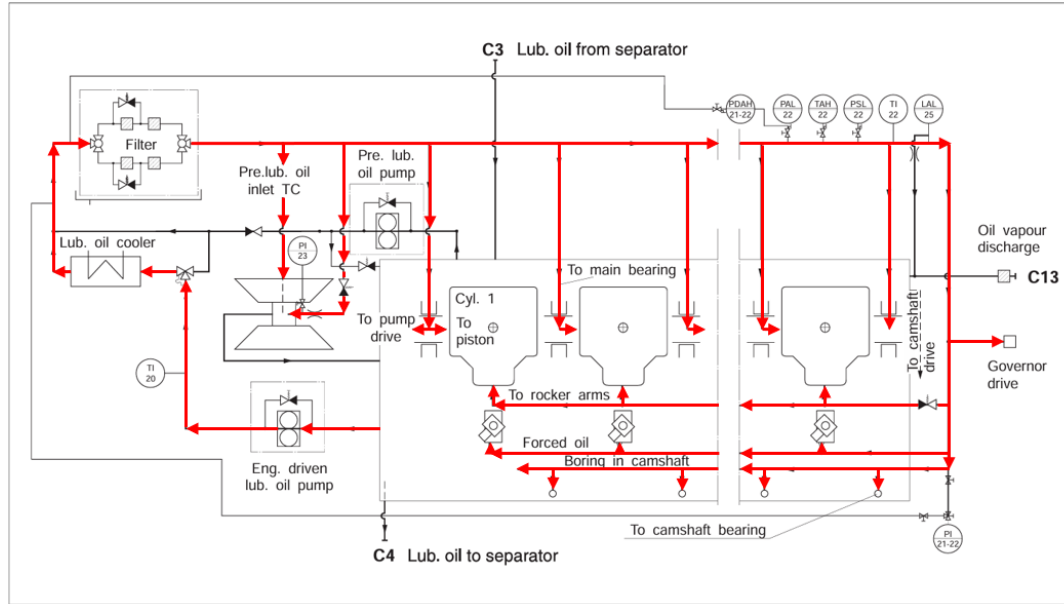
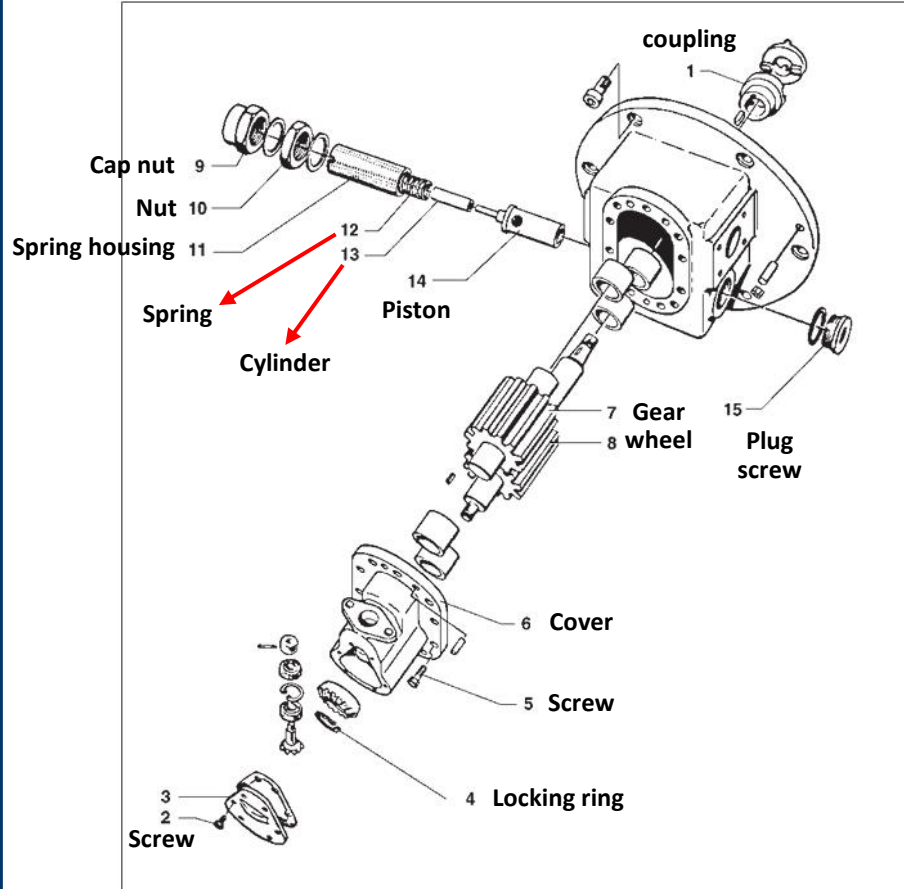


Fig. 1. Diagram for Internal Lubricating Oil System

The lubricating oil pipes, for the camshaft drive gear wheels, are equipped with nozzles which are adjusted to apply the oil at the points where the gear wheels are in mesh.

The lubricating oil pipe, and the gear wheels for the governor drive are adjusted to apply the oil at the points where the gear wheels are in mesh.

Engine driven lubricating oil pump

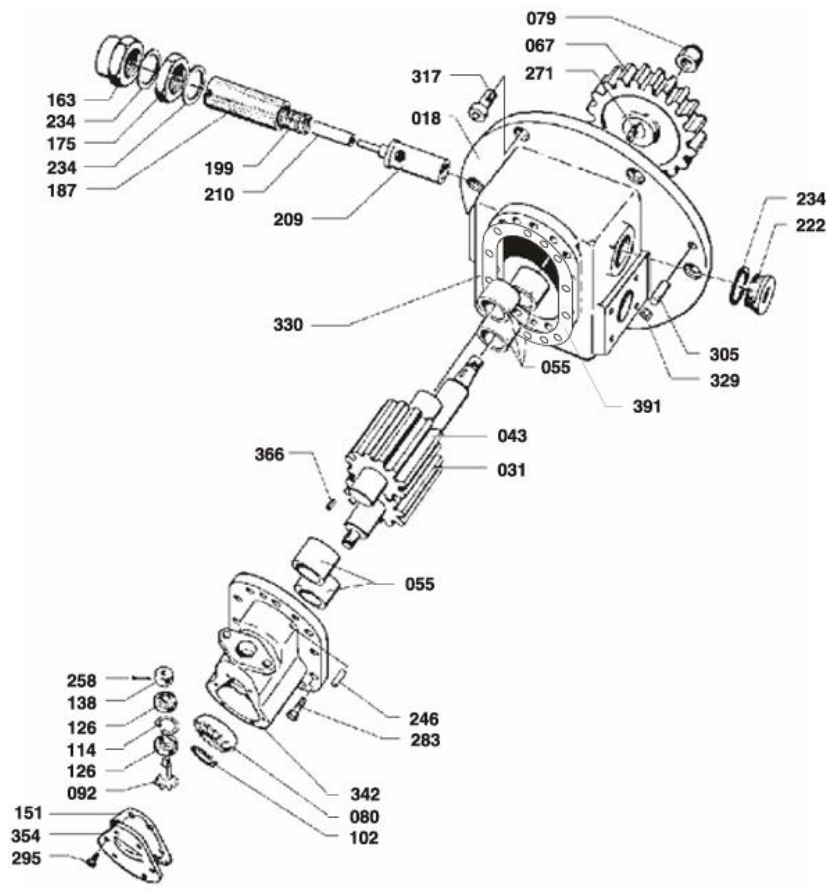


The lubricating oil pump, which is of the gear wheel type, is mounted on the front end of the engine and is driven by means of the crankshaft through a coupling or a gear wheel.

The pressure regulator forms part of the lubricating oil pump. Adjustment of the oil pressure is done by removing the cap nut and turning the adjusting screw.



Engine driven lubricating oil pump



Item No.	Qty.	Designation	Benævnelse	Item No.	Qty.	Designation	Benævnelse
018	1/P	Pump housing	Pumpehus	317	6/P	Screw	Skrue
031	1/P	Shaft with gear wheel, long	Aksel med tandhjul, lang	329	2/P	Nut	Motrik
043	1/P	Shaft with gear wheel, short	Aksel med tandhjul, kort	330	1/E	Lub. oil pump, complete	Smøreliepumpe, komplet
055	5/P	Bush	Bøsning	342	1/P	End cover	Endedæksel
067	1/P	Gear wheel	Tandhjul	354	1/P	Cover	Dæksel
079	1/P	Nut	Motrik	366	1/P	Key	Feder
080	1/P	Cog wheel	Konisk tandhjul	391	1/P	Gasket	Pakning
092	1/P	Cog wheel	Konisk tandhjul				
102	1/P	Circlip	Sikringsring				
114	1/P	Circlip	Sikringsring				
126	2/P	Ball bearing	Kugleleje				
138	1/P	Socket	Muffe				
151	1/P	Gasket	Pakning				
163	1/P	Cap nut	Hættetotrik				
175	1/P	Nut	Motrik				
187	1/P	Adjusting screw	Justerskrue				
199	1/P	Spring	Fjeder				
209	1/P	Piston	Stempel				
210	1/P	Cylinder	Propskrue				
222	1/P	Plug screw	Pakning				
234	3/P	Gasket	Cylindrisk stift				
246	2/P	Cylindrical pin	Stift				
258	1/P	Pin	Feder				
271	1/P	Key	Skrue				
283	10/P	Screw	Skrue				
295	6/P	Screw	Cylinder				
305	2/P	Guide pin	Styrestift				

Pre lubricating pump

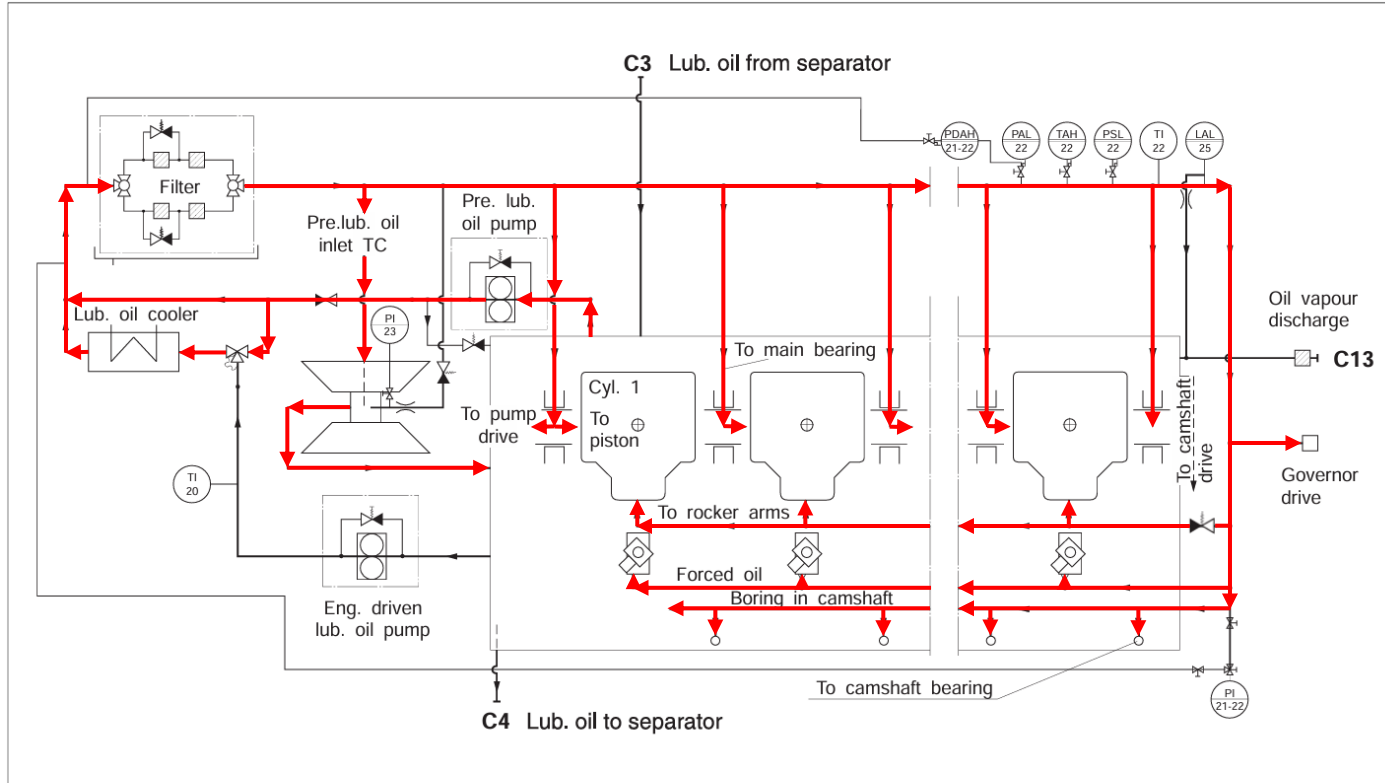


Fig. 1. Diagram for Internal Lubricating Oil System

Pre lubricating pump

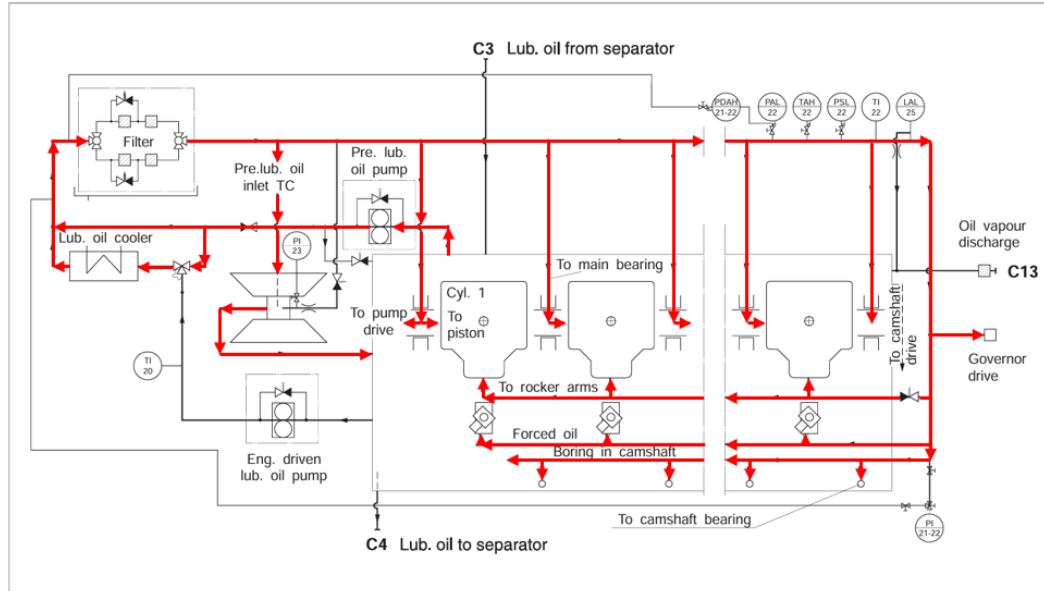
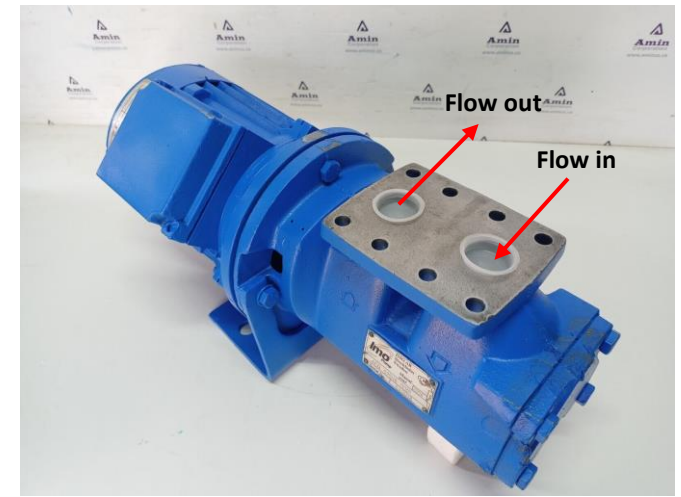


Fig. 1. Diagram for Internal Lubricating Oil System



As standard, the engine is equipped with an electric driven pre lub. pump mounted parallel to the main pump.

The pump must be arranged for automatic operation, ensuring stand-still of the pre lubricating pump when the engine is running, and running during engine stand-still in stand-by position.

Pre lubricating pump

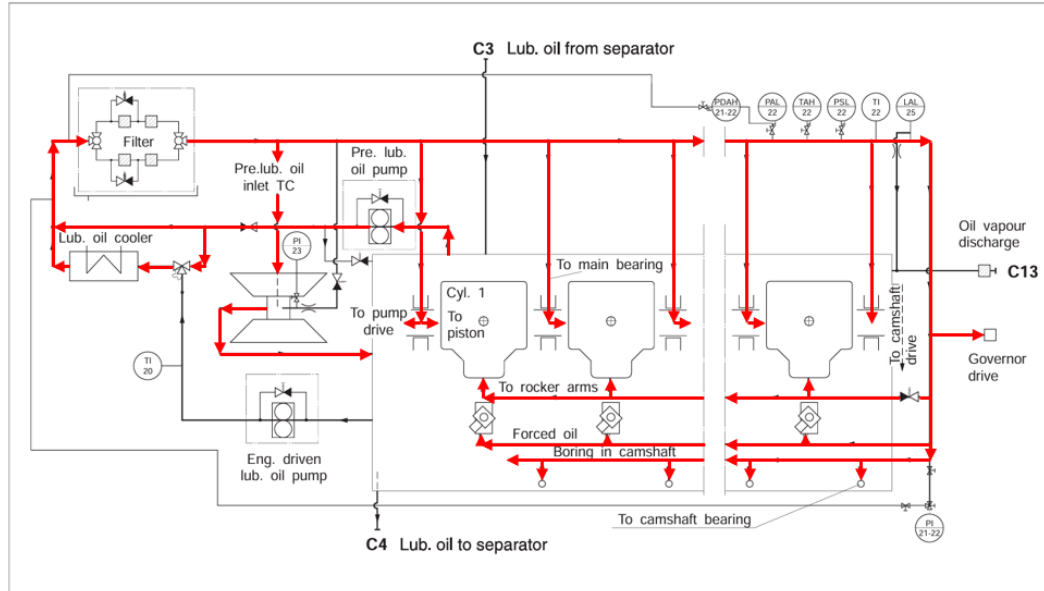


Fig. 1. Diagram for Internal Lubricating Oil System

Running period of the pre lubricating pump is preferably to be continuous. If intermittent running is required for energy saving purpose, the timing equipment should be set for shortest possible intervals, say 2 minutes of running, 10 minutes of stand-still, etc.

Further, it is recommended that the pre lub. pump is led from the emergency switch board thus securing that the engine is not started without pre lubrication.

Pre lubricating pump

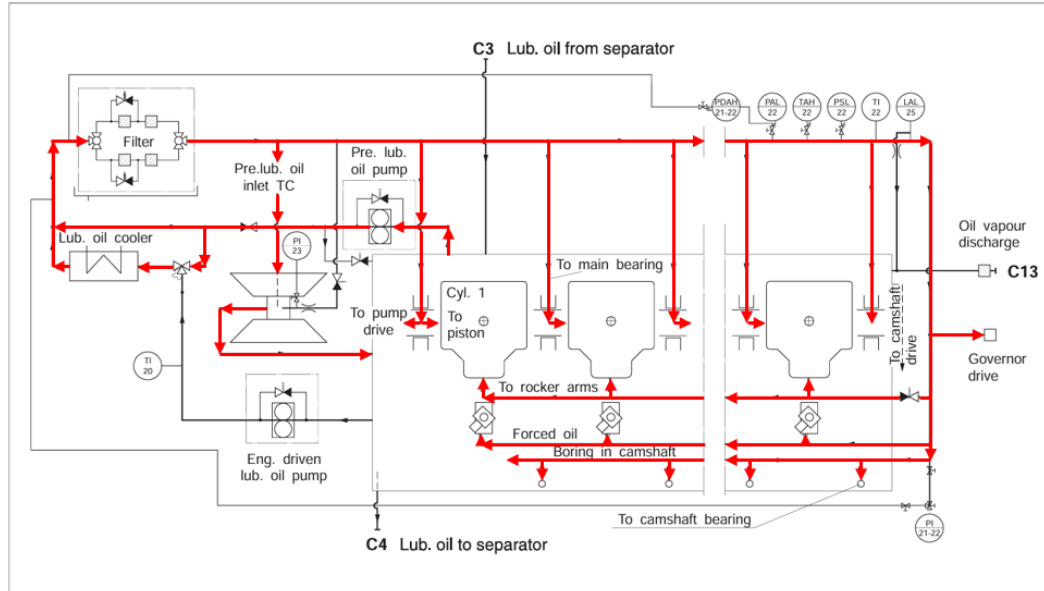


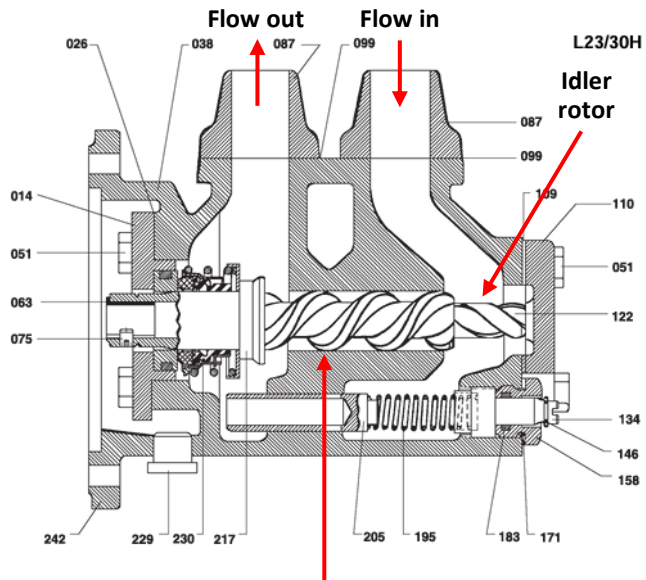
Fig. 1. Diagram for Internal Lubricating Oil System

The pre-lubricating pump must always be operated while the G/E is stopped to supply lubricating oil to each part of the G/E.

If the G/E is started without the pre-lubricating pump running, the crankshaft and bearings (connecting rod bearings and main bearings) will suffer severe damage, making it impossible to operate the G/E any longer.

2nd marine engineer must check that the L.O pressure of G/E is not zero (due to running of pre-lubricating pump) before starting G/E.

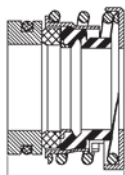
Pre lubricating pump



Power rotor

L23/30H

Idler rotor



230

Item No.	Qty.	Designation
014	1/P	Front cover
026	1/P	Joint
038	1/P	Pump casing
051	8/P	Screw
063	1/P	Socket
075	1/P	Set screw
087	2/P	Counter flange, complete
099	2/P	Joint
109	1/P	Joint
110	1/P	Rear cover
122	2/P	Idler rotor, only available together with item 217
134	1/P	Regulating screw
146	1/P	Retaining ring
158	1/P	Valve cover
171	1/P	Sealing washer
183	1/P	O-ring
195	1/P	Valve spring
205	1/P	Valve piston
217	1/P	Power rotor, only available together with item 122
229	1/P	Drip ring
230	1/P	Shaft seal, complete
242	1/E	Prelubricating pump, complete

Exploded view

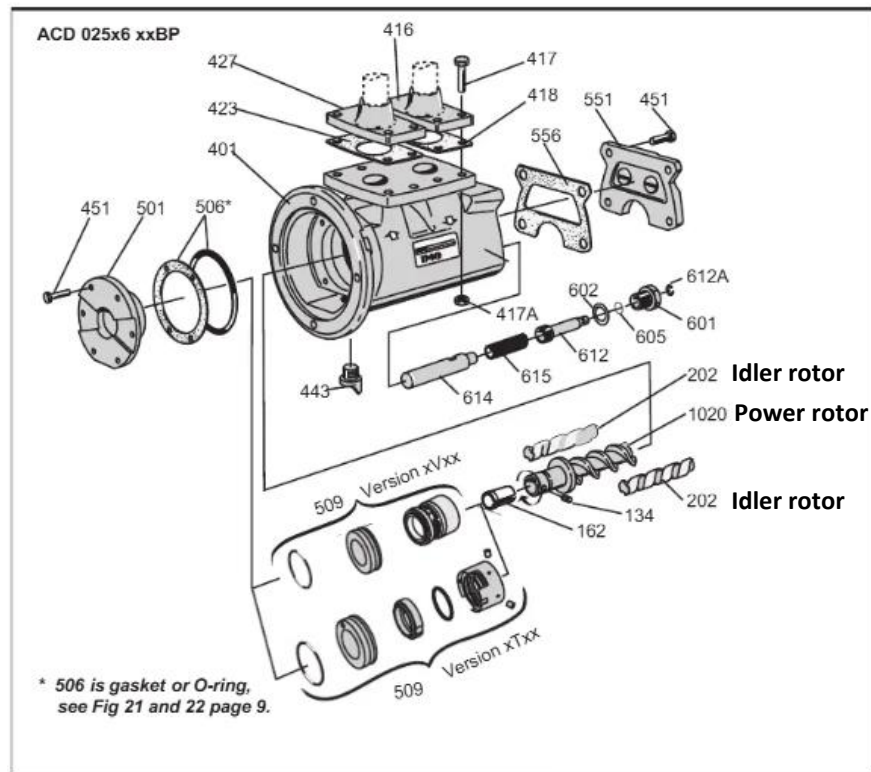


Fig. 1

Pre lubricating pump

Driving Screw (Power Rotor)

A central screw that rotates directly connected to a motor.

Driven Screws (Idler Rotors)

Two auxiliary screws that mesh with the driving screw and rotate together.

Casing

Encloses the screws and forms a fluid flow path.

Exploded view

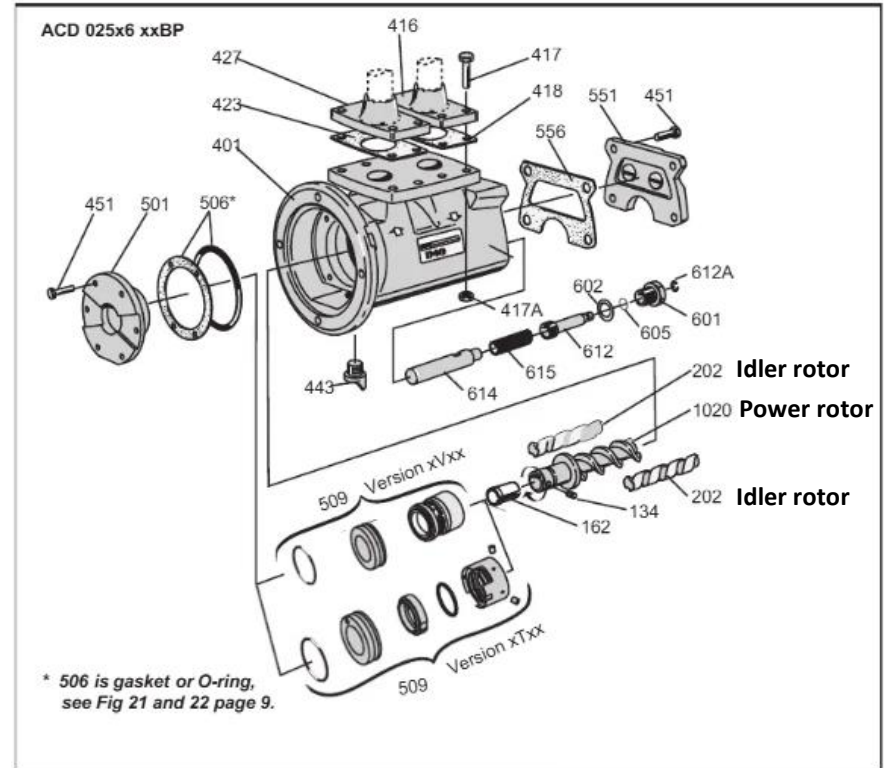
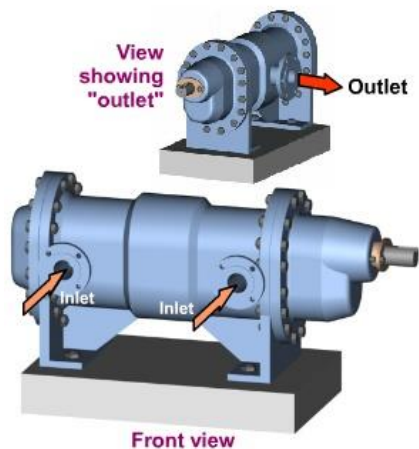


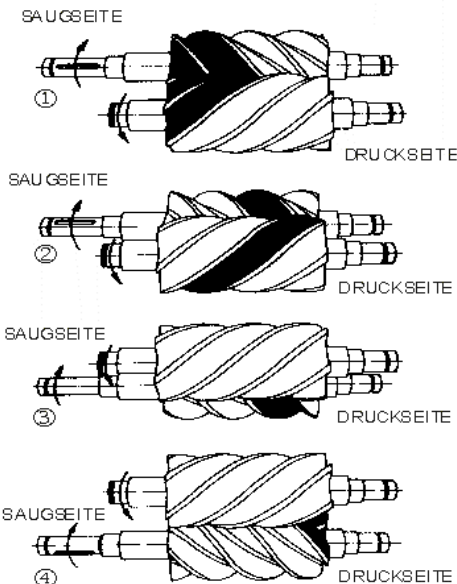
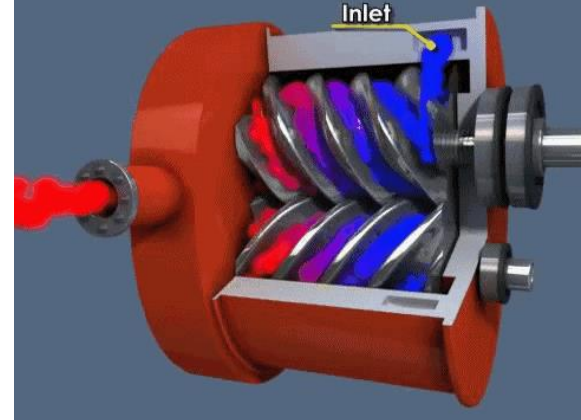
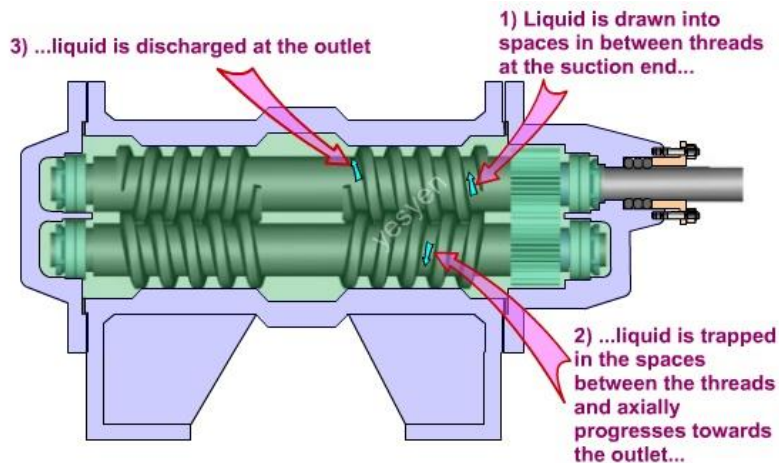
Fig. 1

Pre lubricating pump

Principle of twin screw pump

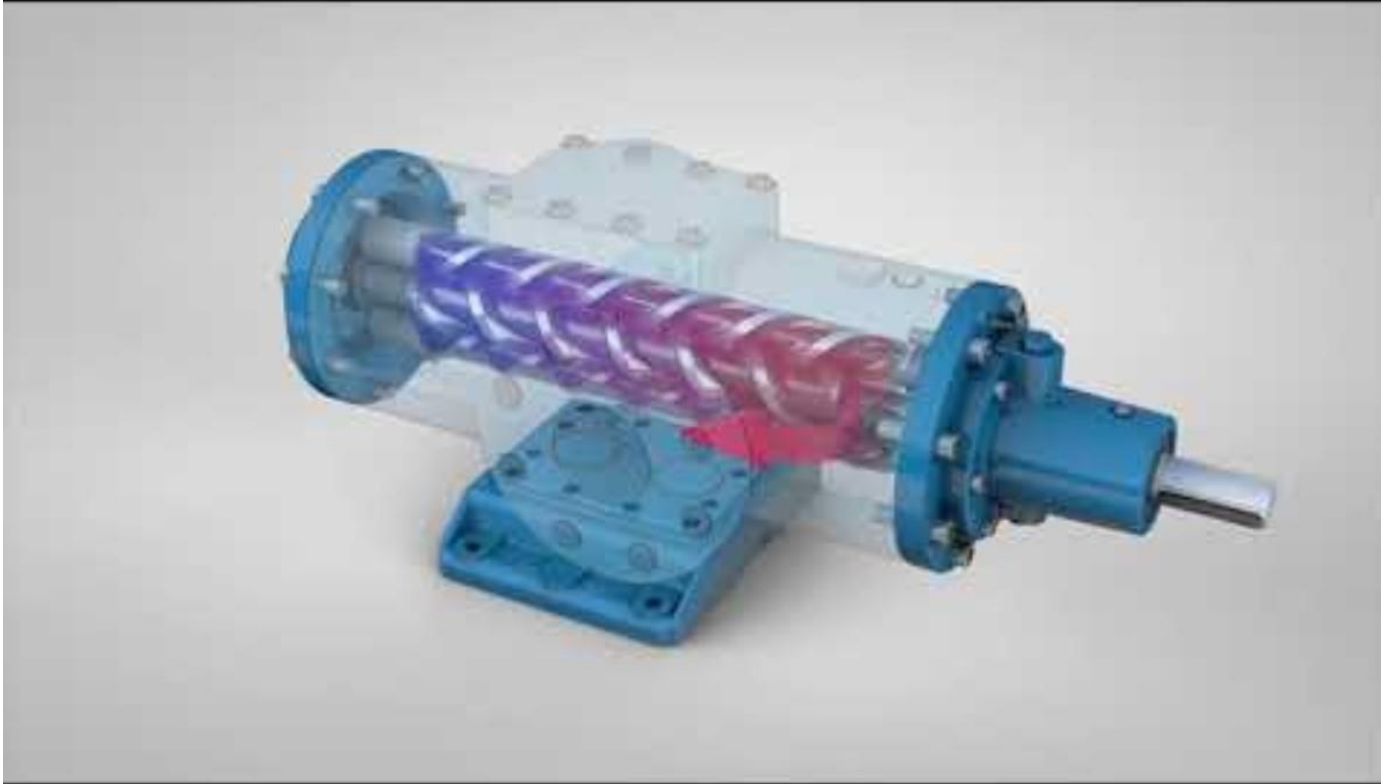


DOUBLE SUCTION - DOUBLE SCREW PUMP - WORKING



Pre lubricating pump

Principle of triple screw pump



Pre lubricating pump

Suction Phase

As the driving screw rotates, the two meshed driven screws also rotate in the opposite direction. This creates a sealed space (pocket) between the screw teeth and the housing wall. As this space expands at the pump inlet, fluid is sucked in. (As the enclosed space expands with rotation, its internal pressure drops, sucking fluid from the pump inlet.)

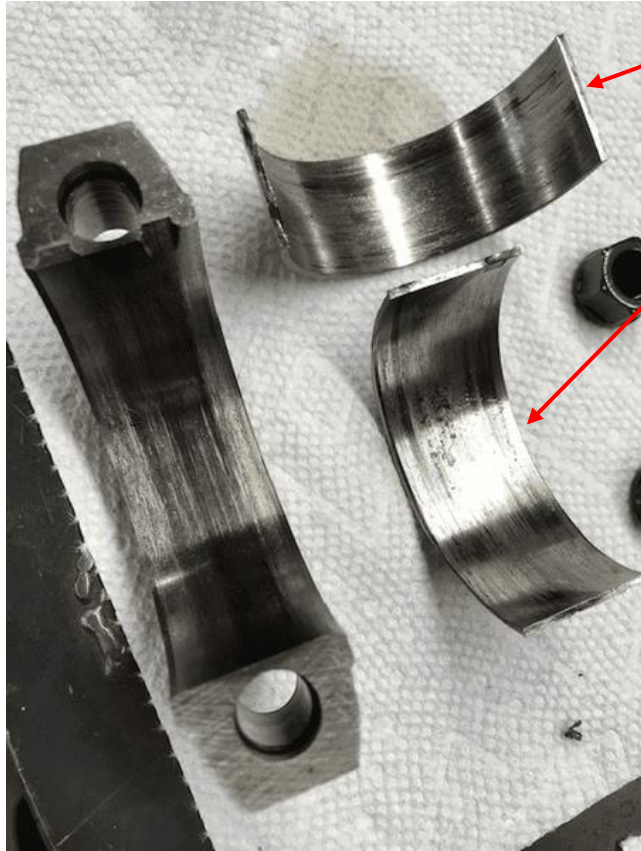
Transport Phase

As the screws rotate, the sealed spaces (pockets) move axially (along the length of the screw). During this process, the fluid is transported toward the outlet at a constant velocity without compression. (At this time, the fluid is pushed out without being compressed and maintaining a nearly constant volume.)

Discharge Phase

As the sealed spaces disappear at the screw tips, the fluid is forced toward the outlet. Since multiple pockets move continuously, the flow rate remains constant with minimal pulsation. (When the confined space reaches the outlet, the fluid inside experiences a pressure buildup because there is no further room for the space to move forward and the outlet is blocked. This pressure buildup ultimately pushes the fluid out with a force greater than the external piping/system pressure.)

Pre lubricating pump



Wear of connecting rod bearing



Wear of crank pin

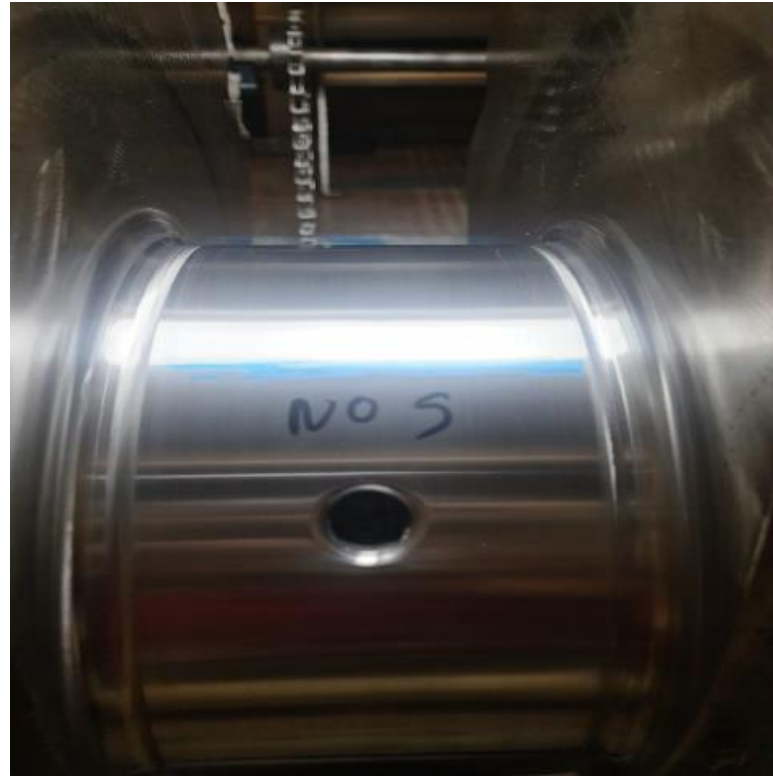


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Pre lubricating pump



Wear of crank pin



The crank pin on the new crankshaft is shiny.



Lubricating Oil Filter

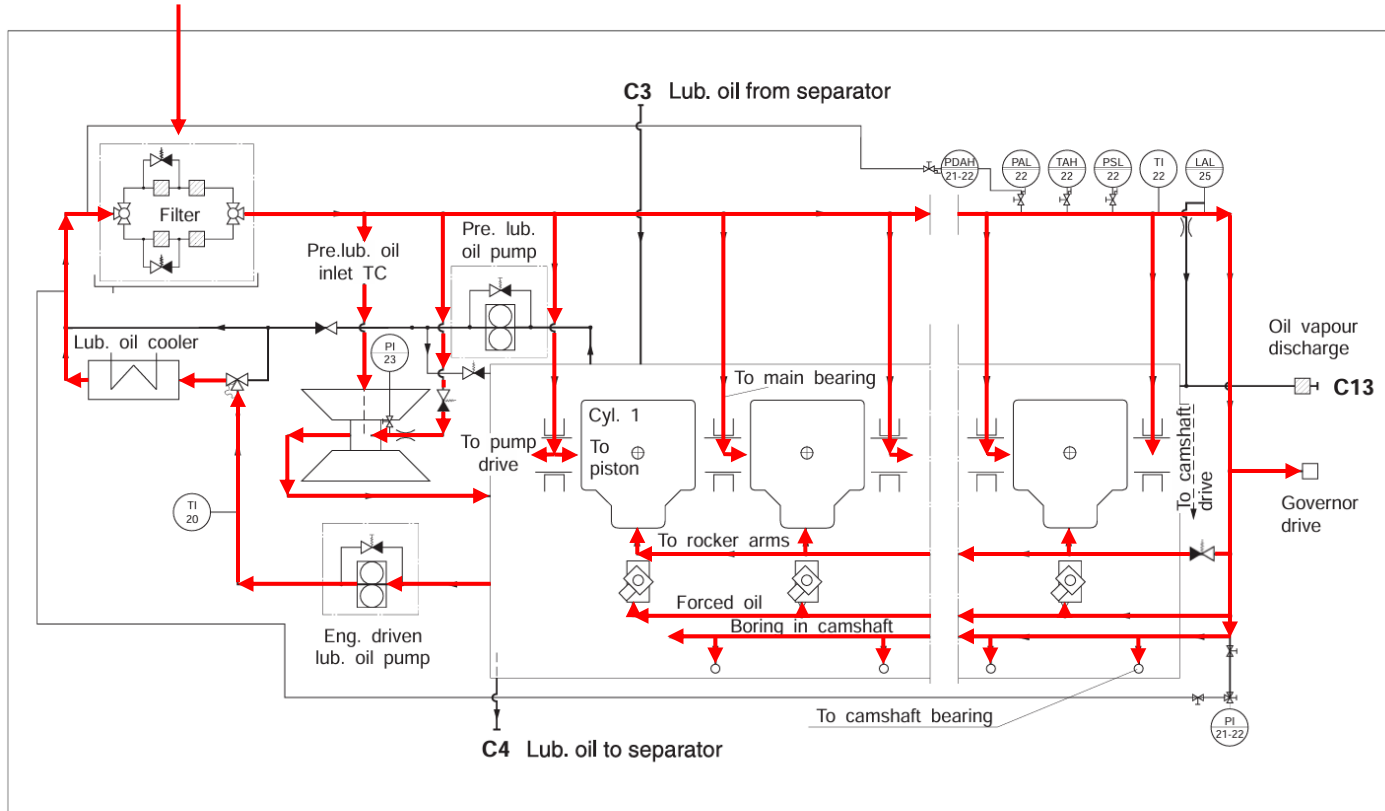


Fig. 1. Diagram for Internal Lubricating Oil System

Lubricating Oil Filter (HiMSEN)



Lubricating Oil Filter

The lubricating oil filter is a double filter which is generally used with only one filter chamber being in operation, the other filter chamber being stand-by.

If the filter chamber in operation needs to be serviced, the operation can be switched to the other filter chamber without any interruption in lubricating oil supply to the engine.

Each filter chamber is equipped with 1 or 2 replaceable paper cartridges of fineness 10-15 microns.

In the center of each filter chamber a filter basket (central element) is situated. This filter basket is acting as a safety filter, having a fineness of about 60 microns.

**Filter basket
(central element)**

Paper cartridges



Lubricating Oil Filter

During operation, an increased pressure drop across the filter will be observed as dirt particles will deposit on the filtration surfaces of the paper cartridges and thus increase the flow resistance through the filter.

To ensure safe filtering of the lubricating oil, none of the by-pass valves must open during normal service and the elements should be replaced at a pressure drop across the filter of 1.5 bar.

When exchanging cartridges, it is advisable to release any old oil remaining in the filter housing by means of the drain plug provided for this purpose, and to wipe out the housing with a cloth.



Centrifugal Oil Filter



Centrifugal Oil Filter



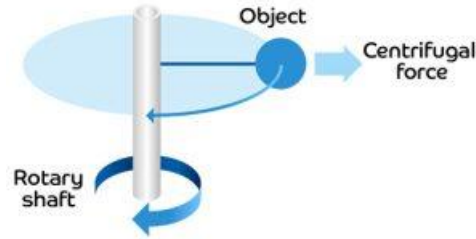
Centrifugal Oil Filter





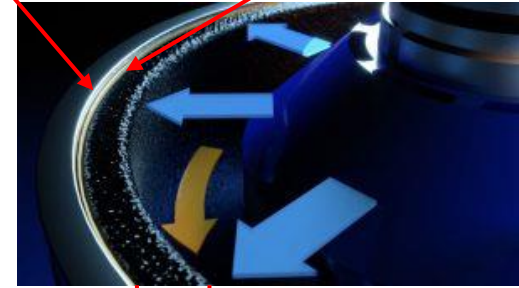
Centrifugal Oil Filter

As the rotor rotates, centrifugal force is applied, causing heavy impurities such as sludge, metal particles, and carbon deposits in the oil to adhere to the rotor walls. (Similar to the principle used in washing machines.)



Cover of the centrifugal oil filter

Paper cartridge filter



Carbon deposits

Accordingly, clean oil returns to the G/E L.O. sump tank by gravity.

To more easily separate the carbon deposits stuck inside the centrifugal oil filter, a thin paper cartridge is installed inside the cover of the centrifugal oil filter.



Centrifugal Oil Filter

Oil Inflow

Oil from the engine enters the filter rotor under pressure.

The oil enters through the rotor's central axis and collects within the rotor's internal space.

Nozzle (Jet) Structure

A very small jet orifice is located on the bottom or side of the rotor.

This nozzle is positioned slightly obliquely from the rotor center.

Jet Reaction (Newton's Third Law)

When oil is rapidly ejected through the nozzle under pressure,
the rotor rotates due to the reaction force of the ejected fluid's momentum.

Continuous Rotation

Since the engine oil pump continuously supplies oil, a continuous jet of oil is ejected from the nozzle.

As a result, the rotor rotates at high speeds (typically over several thousand RPM).



Centrifugal Oil Filter



L.O. transfer & purifying system for G/E

When the L.O of the G/E is severely contaminated, the L.O purifier can be operated to purify the L.O in the G/E L.O sump tank.

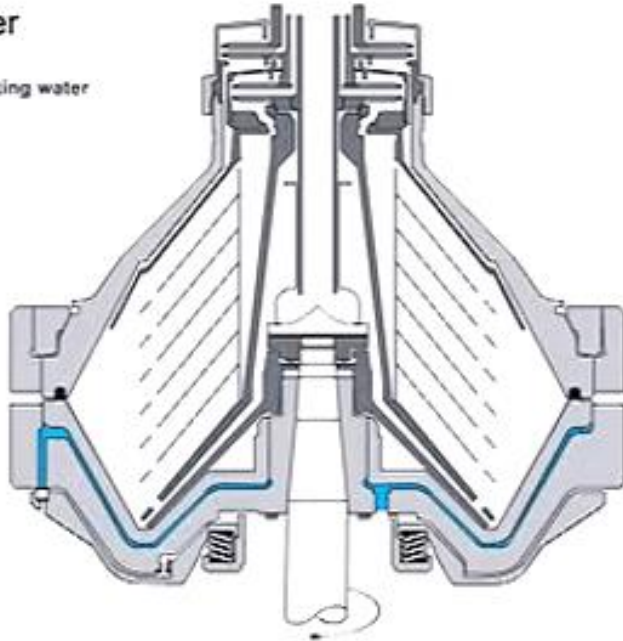
The L.O purifier can be operated continuously regardless of whether the G/E is in operation.

When the L.O purifier is in operation, it is necessary to consider the case where water is mixed into the purified L.O and goes to the G/E L.O sump tank, and the case where the L.O to be purified leaks from the L.O purifier and the amount of L.O in the G/E L.O sump tank decreases.

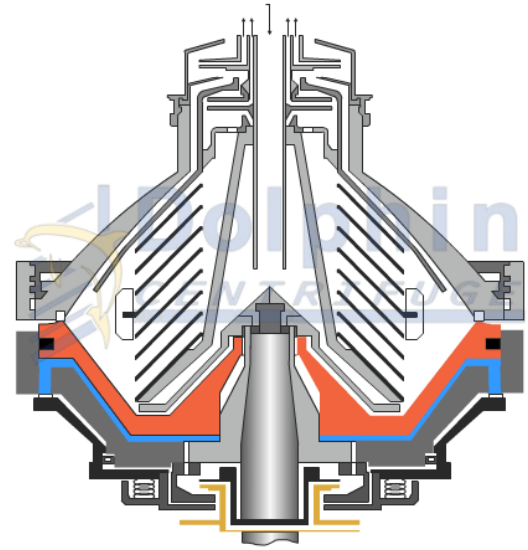
L.O. purifier

Purifier

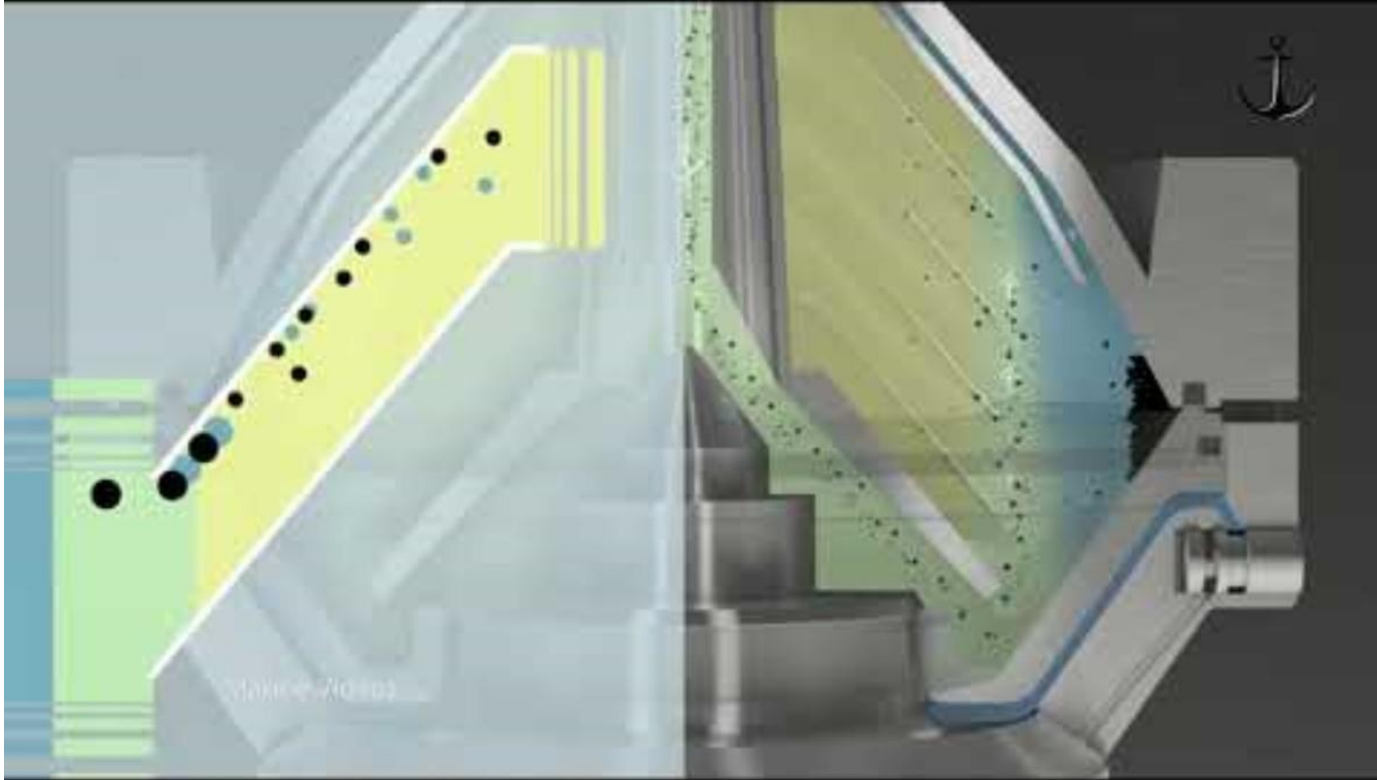
■ Operating water



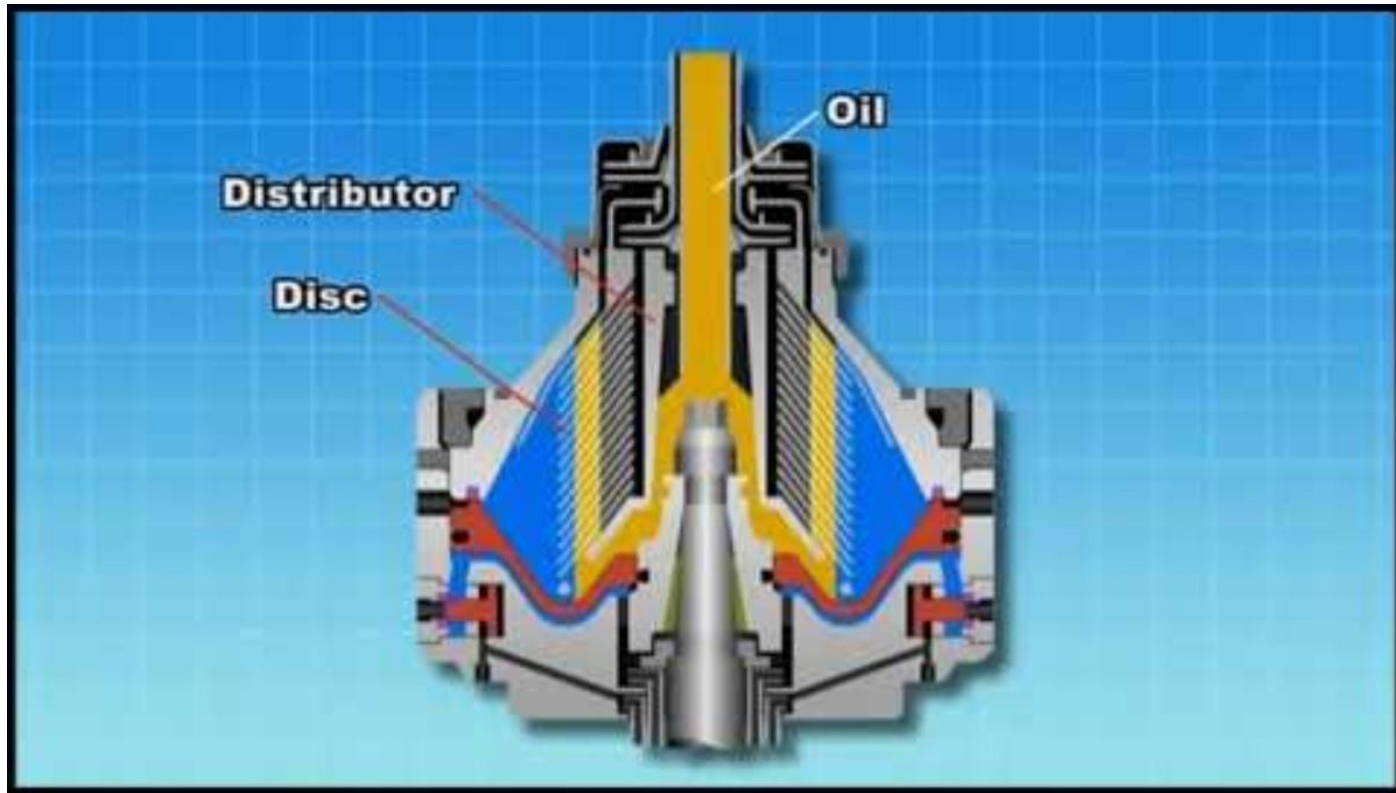
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L.O. purifier



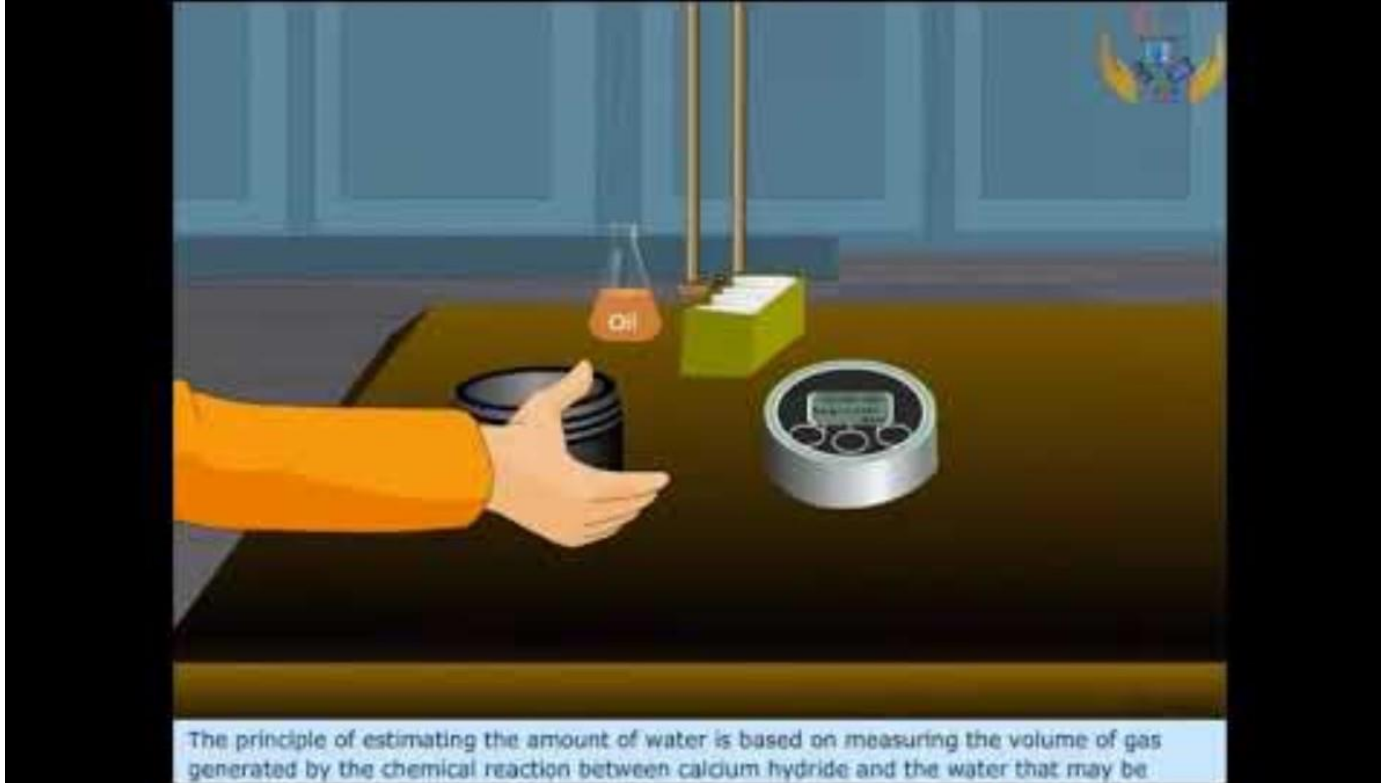
L.O. purifier



L.O. purifier



L.O. test onboard



L.O. test onboard



L.O. test onboard



L.O. test onboard



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